

Keep Your Kubernetes Secrets Safe in Git

Keeping passwords out of Git is one of the most important habits to build as a cloud engineer. The Helm Secrets plugin lets you safely encrypt your sensitive values so you can track them in Git without leaking them to the world.

What you'll learn:

- 1 Installing Helm Secrets & Age
- 2 Encrypting Your First Secrets File
- 3 Editing Encrypted Files Safely
- 4 Deploying to Kubernetes

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Installing Helm Secrets & Age

Before you can encrypt anything, you need two lightweight tools: the Helm Secrets plugin and Age (a modern tool to generate encryption keys). This setup takes less than two minutes and works entirely on your local machine without needing a complex cloud setup.

```
$helm plugin install https://github.com/jkroepke/helm-secrets  
age-keygen -o key.txt
```

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Encrypting Your First Secrets File

Create a file called `secrets.yaml` with your database passwords. Then, use Sops (which runs under Helm Secrets) to encrypt it using the key you generated earlier. This turns your plain-text passwords into scrambled, safe text.

```
$export SOPS_AGE_KEY_FILE=$(pwd)/key.txt
helm secrets encrypt -i secrets.yaml
```

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Editing Encrypted Files Safely

If you need to change a password, you do not have to decrypt the file, edit it, and encrypt it again. Use the edit command to safely open the file in your terminal's editor, make your changes, and let Helm Secrets automatically re-encrypt it when you save and close.

```
$export SOPS_AGE_KEY_FILE=$(pwd)/key.txt  
helm secrets edit secrets.yaml
```

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Deploying to Kubernetes

When you are ready to deploy, Helm Secrets decrypts the file in your computer's memory just long enough to send the values to Kubernetes, then discards the plain text. Your passwords are never written to your hard drive or Git repository in plain text during this process.

```
$export SOPS_AGE_KEY_FILE=$(pwd)/key.txt  
helm secrets upgrade --install my-app ./my-chart -f values.yaml -f secrets.yaml
```

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Key Takeaways

- + Add your private key files (like key.txt) to your .gitignore file before doing anything else.
- + Store your decryption keys in a secure password manager or a cloud secret manager (like AWS Secrets Manager).
- + Name your encrypted files clearly (e.g., secrets.yaml) so your team knows they contain sensitive data.
- + Use read-only access for keys in your CI/CD pipelines to prevent accidental modifications during deployment.
- + Run 'helm secrets view' to double-check your configurations without risking accidental changes.

Watch Out For

- ! Forgetting to set the SOPS_AGE_KEY_FILE environment variable, which causes decryption commands to fail.
- ! Committing the raw, unencrypted backup files to Git by mistake.
- ! Sharing the private age key over public channels like Slack or email instead of a secure vault.

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